

Peak-Aware Cost-Volume Learning for Multichannel Bioacoustic Alignment

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Abstract—Temporal misalignment is common in multichannel bioacoustic analysis, especially when recordings are acquired using independent devices or unsynchronized channels. We formulate alignment as the estimation of a time-varying inter-channel delay trajectory and adapt cost-volume matching to this setting. The peak-aware cost-volume model constructs explicit matching evidence over candidate temporal lags from encoded two-channel short-time Fourier transform magnitude features. Peak-aware pooling then summarizes the lag-domain evidence into dominant responses and reliability cues, while a bidirectional long short-term memory models their temporal evolution. The estimated frame-level delay trajectory is then interpolated to the target keypoints and converted into matching timestamps in the other channel. On the zebra finch validation subset, the neural model improves coarse alignment over the official deep-learning baseline. In the final submission, the neural estimator is combined with generalized cross-correlation with phase transform (GCC-PHAT) through confidence-based routing, with the GCC-PHAT branch providing the fine-alignment gains on aru.

Index Terms—multichannel bioacoustic alignment, cost volume, peak-aware pooling, time-delay estimation, GCC-PHAT

1. INTRODUCTION

Reliable temporal correspondence across recording channels is essential for comparing, fusing, and jointly analyzing bioacoustic events. Recent studies have used passive recordings to quantify porpoise detection probability, detect weak marine signals, denoise odontocete clicks, and track echolocating whales [1]–[4]. In practice, independently clocked recorders introduce unknown timing offsets and gradual clock drift, while noise, overlapping sounds, and intermittent shared vocal activity make local cross-channel correspondence difficult to recover. These challenges motivate alignment methods that can track temporal variation rather than assume a single global shift.

Classical approaches to time-delay estimation commonly rely on cross-correlation or generalized cross-correlation (GCC) [5]. Entropy-based criteria, adaptive filtering, and relative impulse-response modeling have also been investigated to improve robustness under noisy and reverberant conditions [6]–[11]. For long bioacoustic recordings, however, correlation-based methods face a fundamental trade-off: a single global-delay estimate cannot capture temporal drift, whereas sliding-window estimates may become unreliable when shared acoustic events are sparse or ambiguous.

Recent neural approaches to time-delay estimation have explored learned GCC-based representations, shift-equivariant preprocessing, correlation-based objectives, and self-supervised cross-channel matching [12]–[16]. These approaches improve the representation and recovery of cross-channel correspondence under challenging acoustic conditions. However, many existing formulations estimate a single delay from a fixed observation window rather than a continuous delay trajectory over a long recording. Time-varying bioacoustic alignment

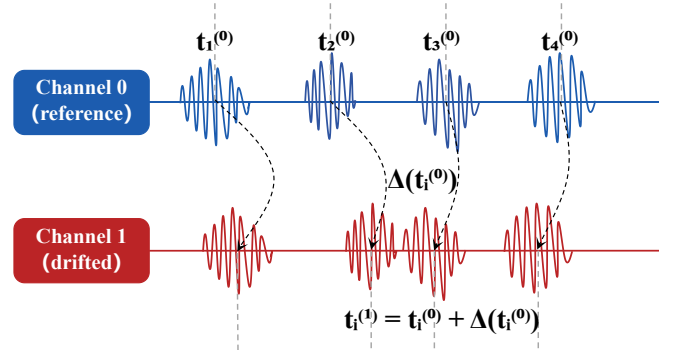


Fig. 1: Illustration of the multichannel alignment task. Channel 0 provides query timestamps, while Channel 1 contains corresponding acoustic events under a time-varying inter-channel delay. Dashed arrows indicate the delays at different query times, and the task is to predict the corresponding Channel 1 timestamp for each Channel 0 query.

therefore requires local lag evidence to be interpreted jointly with temporal context.

BioDCASE 2026 Task 1 provides a benchmark for this setting. Given timestamps in Channel 0, a system must predict the matching timestamps in Channel 1. The task includes a fine-grained setting, in which millisecond errors matter, and a coarse setting with larger desynchronization. Together, these settings require both millisecond-scale precision and robustness to larger, time-varying misalignment.

We formulate alignment as frame-level estimation of a time-varying inter-channel delay trajectory. The neural branch adapts group-wise correlation and correlation-and-difference cost construction [17], [18] to lag-time matching of encoded two-channel short-time Fourier transform (STFT) magnitudes, and combines peak-aware pooling with a bidirectional long short-term memory (BLSTM) [19] to estimate the trajectory. The final submission routes between the neural estimator and a GCC with phase transform (GCC-PHAT) branch according to confidence. The main contributions are:

- an adaptation of cost-volume matching to frame-level delay-trajectory estimation for bioacoustic timestamp alignment;
- a combination of peak-aware lag pooling and temporal modeling, evaluated through neural ablations on the aru and zebra finch subsets;
- a confidence-routed hybrid system that ranked first in the official BioDCASE 2026 Task 1 evaluation, with validation results showing neural gains on coarse alignment and GCC-PHAT gains on fine alignment.

2. PROBLEM FORMULATION

We view multichannel alignment as estimating a time mapping between two recordings of the same acoustic scene. Let $x_0[n]$ and $x_1[n]$ denote

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the reference and drifted channels. If an acoustic event is observed at time t in Channel 0, its matching time in Channel 1 can be written as

$$\phi(t) = t + \Delta(t), \quad (1)$$

where $\phi(t)$ is the cross-channel time mapping and $\Delta(t)$ is the relative delay. Unlike a constant-offset model, $\Delta(t)$ can vary over time, thereby capturing clock drift and local timing variations. The alignment problem is to estimate $\Delta(t)$ from the two-channel audio, from which $\phi(t)$ follows directly.

In the evaluation setting used in this paper, predictions are required only at a finite set of Channel 0 query times. For each recording,

$$T^{(0)} = \{t_i^{(0)}\}_{i=1}^N, \quad (2)$$

where N denotes the number of query times. The corresponding ground-truth Channel 1 timestamps satisfy

$$t_i^{(1)} = t_i^{(0)} + \Delta(t_i^{(0)}). \quad (3)$$

Given an estimated delay trajectory $\hat{\Delta}(t)$, the predicted timestamps are

$$\hat{t}_i^{(1)} = t_i^{(0)} + \hat{\Delta}(t_i^{(0)}). \quad (4)$$

The formulation separates the continuous alignment problem from the discrete keypoint output format. Performance is measured using mean absolute error (MAE) in milliseconds and mean squared error (MSE) in squared seconds.

3. METHOD

3.1. Overview

The neural branch estimates a delay trajectory instead of predicting each timestamp independently. Given a two-channel recording and Channel 0 query timestamps, it computes STFT magnitude features and maps both channels through a shared frequency-time encoder. The encoded features are compared over candidate lags to form a lag-time cost volume. Lag-time refinement, peak-aware pooling, and a BLSTM head then produce the neural delay trajectory $\hat{\Delta}_{nn}(t)$ at each frame. Channel 1 timestamps are obtained by interpolating this trajectory at the query times and adding the estimated delay to each Channel 0 timestamp. In the hybrid system, confidence-based routing assigns high-confidence recordings to GCC-PHAT and the remaining recordings to the neural branch.

3.2. Input representation and target construction

All audio is resampled to 16 kHz. For the neural model, we compute two-channel STFT magnitudes using a 2048-sample Hann window and an 800-sample (50 ms) hop. The direct-current bin is removed, yielding an input tensor of size $2 \times 1024 \times T$. The network applies logarithmic compression and layer normalization across the channel and frequency dimensions. For supervision, ground-truth timestamp pairs are converted into frame-level delay targets in seconds on the same 50 ms temporal grid. The target sequence begins 1 s after recording onset, and a validity mask excludes frames without training targets from the loss.

3.3. Weight-shared frequency-time encoder

The two channels are processed independently by a weight-shared encoder before lag matching. Weight sharing provides a common feature space for cross-channel comparison, while frequency-axis compression aggregates spectral information without reducing temporal resolution. The submitted neural model uses eight encoder blocks with output channel widths of 16, 32, 64, 128, 128, 128, 128, and 128. Each block uses a two-dimensional convolution with kernel size 5×5 , a

frequency stride of 2, layer normalization, parametric rectified linear unit activation, and a temporal depthwise separable convolution with kernel size 5. The encoder reduces the frequency axis from 1024 bins to 4 bins while preserving the temporal resolution for lag-domain comparison.

3.4. Lag-time cost volume

Rather than regressing delay directly from the encoded features, the model compares the encoded channels over candidate lags. The cost volume spans delays from -6 s to 6 s. With an STFT hop of 50 ms and a two-frame lag step, adjacent coarse lag candidates are separated by 100 ms. We therefore define $\mathcal{L} = \{-120, -118, \dots, 118, 120\}$ as the set of candidate frame offsets. For notational simplicity, the four residual frequency bins are absorbed into the feature index c . Following group-wise correlation for cost-volume construction [17], the matching score at a candidate lag $\ell \in \mathcal{L}$ is computed between the encoded Channel 0 representation $\mathbf{F}_0$ and shifted Channel 1 representation $\mathbf{F}_1$:

$$C_g(\ell, t) = \frac{1}{|g|} \sum_{c \in g} \mathbf{F}_{0,c}(t) \mathbf{F}_{1,c}(t + \ell), \quad (5)$$

where g indexes a channel group, c indexes an encoded feature coefficient, and ℓ denotes an integer frame offset.

Following the complementary correlation-and-difference cost construction of Park and Lee [18], the model also computes a negative group-wise absolute-difference score

$$A_g(\ell, t) = -\frac{1}{|g|} \sum_{c \in g} |\mathbf{F}_{0,c}(t) - \mathbf{F}_{1,c}(t + \ell)|. \quad (6)$$

The correlation and difference terms provide complementary similarity and mismatch evidence on the same lag grid. With eight groups, their concatenation gives 16 matching channels. Lag-wise centering removes lag-independent score offsets, after which normalized signed-lag and absolute-lag coordinates are appended, giving 18 input channels. A cost-volume stem maps the resulting volume to 64 hidden channels, followed by two residual lag-time convolutional blocks and two gated refinement layers.

3.5. Peak-aware lag pooling and temporal regression

At each frame, the refined cost volume retains several plausible delays. Peak-aware pooling summarizes both the dominant lag response and its reliability. Separate coarse and fine modules learn scalar lag scores and pool the corresponding responses. Before fine pooling, the lag axis is upsampled by a factor of two, giving a fine lag spacing of about 50 ms. Each module returns a pooled feature together with descriptors of expected lag, spread, entropy, peak contrast, and peak probability. The coarse and fine summaries are concatenated and projected to a 128-dimensional frame-level sequence.

Let $\mathbf{H}(\ell, t)$ denote the refined cost-volume feature at lag ℓ and frame t . The pooling module first predicts a scalar logit $s(\ell, t)$ and normalizes it along the lag axis:

$$p(\ell, t) = \frac{\exp(s(\ell, t)/\tau)}{\sum_{\ell' \in \mathcal{L}} \exp(s(\ell', t)/\tau)}, \quad (7)$$

where τ is a learnable temperature constrained to a stable range. The attentive feature $\sum_{\ell} p(\ell, t) \mathbf{H}(\ell, t)$ is concatenated with the mean, maximum, and standard deviation over the lag axis. Additional descriptors are the expected normalized lag, lag spread, entropy, top-2 peak contrast, and maximum peak probability. Together, these features provide the BLSTM with local matching evidence and peak-reliability cues.

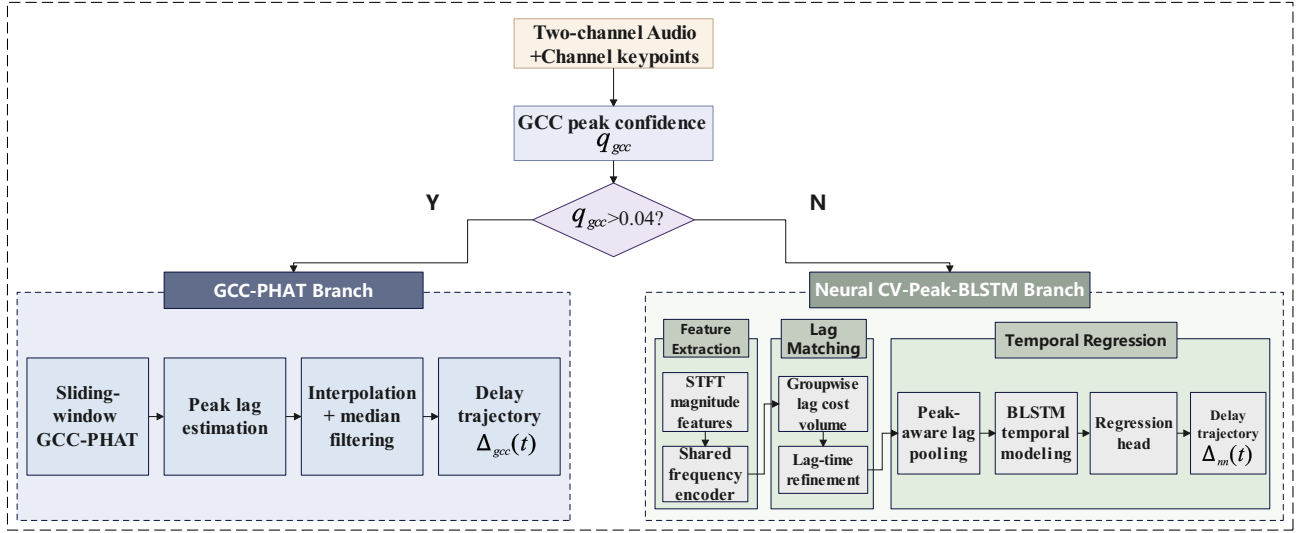


Fig. 2: Overview of the final submitted framework. The neural cost-volume peak-aware (CV-Peak-BLSTM) branch encodes STFT features, constructs and refines a lag-time cost volume, applies peak-aware pooling, and estimates the delay trajectory using a BLSTM head. The submitted system further combines this branch with GCC-PHAT through confidence-based routing.

A two-layer BLSTM with hidden size 128 in each direction models the temporal evolution of the delay trajectory. Its output is projected to 128 dimensions and fused with the local pooled feature using a gated residual context module. A tanh-bounded multilayer perceptron (MLP) maps the fused feature to the frame-level delay estimate $\hat{\Delta}(t)$ within $[-6, 6]$ s. The network has about 3.14 million trainable parameters.

4. EXPERIMENTAL SETUP

4.1. Neural training and inference

The neural model is trained on the official BioDCASE 2026 Task 1 development data without using external data. In our split, the aru subset contains 36 training recordings and 12 validation recordings, while the zebra finch subset contains 108 training recordings and 16 validation recordings. The aru subset represents a fine-grained setting with small temporal offsets, whereas the zebra finch subset represents a coarse setting with larger desynchronization and sparser shared vocal activity. We report MAE and MSE for both subsets and use the unweighted mean across subsets for each metric.

The final fine-tuned model is initialized from an earlier checkpoint, with only the model weights restored and the optimizer and scheduler reinitialized. Training uses masked MAE with an auxiliary masked MSE term over valid frame-level delay targets. The optimized loss is

$$\mathcal{L} = \mathcal{L}_{\text{MAE}} + 0.1 \mathcal{L}_{\text{MSE}}, \quad (8)$$

where both loss terms are averaged over the frames indicated as valid by the target mask $M_{b,t}$, and b and t denote the recording and frame indices, respectively. Fine-tuning uses a batch size of 1, gradient accumulation over 16 steps, AdamW [20] with a learning rate of 10^{-4} , and a reduce-on-plateau learning-rate scheduler.

At inference, the frame-level delay sequence is linearly interpolated at the Channel 0 query times and converted to Channel 1 timestamps using Eq. (4). The hybrid systems apply the same interpolation and timestamp conversion after branch selection.

4.2. Confidence-routed inference

The final submission combines the neural and GCC-PHAT estimators through recording-level confidence routing, as shown in Fig. 2. For each recording, a lightweight GCC-PHAT estimator [5] produces a

delay trajectory $\hat{\Delta}_{\text{gcc}}(t)$ when the mean cross-channel peak confidence is high; otherwise, the system uses the neural estimate $\hat{\Delta}_{\text{nn}}(t)$. The routing rule is

$$\hat{\Delta}(t) = \begin{cases} \hat{\Delta}_{\text{gcc}}(t), & q_{\text{gcc}} > 0.04, \\ \hat{\Delta}_{\text{nn}}(t), & q_{\text{gcc}} \leq 0.04, \end{cases} \quad (9)$$

where q_{gcc} is a recording-level GCC-PHAT confidence. It is computed by averaging the maximum GCC-PHAT values from non-silent 1 s Hann windows with a 50 ms hop, where each maximum is taken over lags within a ± 1 s range. The same branch is therefore used for all query times within a recording.

For the GCC-PHAT branch, a coarse sliding-window delay curve is first interpolated at the Channel 0 query times and then locally refined using 1.5 s windows. The estimate $\hat{\Delta}_{\text{nn}}(t)$ is obtained by interpolating the frame-level output of the CV-Peak-BLSTM branch. We report both the standalone neural estimator and the hybrid variants, while the analysis emphasizes the learned lag-domain branch.

4.3. Implementation and post-processing details

For the GCC-PHAT branch, the coarse stage uses 1 s Hann windows with a 20 ms hop to estimate a local delay curve. The refinement stage re-centers a local window around each coarse estimate and searches a small offset range. Within each candidate window, the residual lag is estimated at the audio sampling resolution. The candidate maximizing a combined peak-strength and peak-sharpness score is selected, with median filtering applied before and after refinement. The post-processing (PP) variant uses the same routing rule but applies additional trajectory-level smoothing after branch estimation.

All reported systems use only the official development data for training and parameter selection. The fixed threshold $q_{\text{gcc}} = 0.04$ is selected on the validation split and held fixed for the official submission.

5. RESULTS AND ANALYSIS

Table 1 compares the standalone neural model with the official deep-learning baseline. On zebra finch, the proposed model reduces MAE from 757.1 ms to 422.9 ms and MSE from 1.218 s^2 to 0.262 s^2 . On aru, the improvement over the official deep-learning baseline is smaller,

Table 1: Validation performance. MAE is reported in ms and MSE in s^2 . Average denotes the unweighted mean across the aru and zebra finch subsets. No sync. denotes no synchronization, DL baseline denotes the official deep-learning baseline, and PP denotes post-processing. GCC-PHAT denotes the provided baseline; the hybrid systems use the locally refined GCC-PHAT branch described in Sec. 4.3.

Method	aru		zebra finch		Average	
	MAE ↓	MSE ↓	MAE ↓	MSE ↓	MAE ↓	MSE ↓
No sync.	116.2	0.019	2297.4	5.734	1206.8	2.877
GCC-PHAT	122.4	0.020	2457.4	7.995	1289.9	4.008
DL baseline	164.3	0.040	757.1	1.218	460.7	0.629
Proposed neural	118.4	0.020	422.9	0.262	270.6	0.141
Hybrid	0.7	8.7e-5	422.9	0.263	211.8	0.131
Hybrid + PP	0.7	8.7e-5	416.1	0.262	208.4	0.131

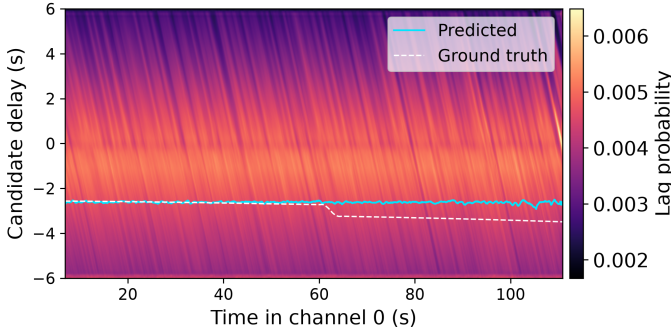


Fig. 3: Representative fine lag-probability map from the zebra finch validation subset. The selected recording has a frame-level delay MAE of 377.7 ms. Brighter regions indicate higher probability for candidate delays; the predicted and ground-truth delay trajectories are overlaid.

and the proposed model remains close to the no-synchronization result, consistent with the smaller temporal offsets in this fine-grained subset. Averaged across the two subsets, the neural model reduces MAE by 41.3% relative to the official baseline. These results are consistent with the benefit of retaining structured candidate-lag evidence before temporal regression, particularly under coarse and time-varying misalignment.

The larger gain on zebra finch is consistent with the coarse-alignment conditions in this subset. Shared calls are sparse, and the correct delay may not produce a sharp local peak at every frame. By preserving multiple lag candidates, the cost volume allows peak-aware pooling and the BLSTM to exploit both local reliability cues and temporal context when estimating the delay trajectory.

The hybrid rows use the same neural architecture as the standalone neural model. The hybrid system’s advantage on aru comes from reliable waveform matching: all 12 validation recordings are routed to GCC-PHAT, where local refinement and median filtering improve delay estimates. On zebra finch, performance remains close to the standalone neural model. The full submitted system, including post-processing, achieves the lowest average validation MAE. Post-processing reduces zebra finch MAE from 422.9 ms to 416.1 ms, indicating a modest benefit from trajectory-level smoothing.

Fig. 3 shows the fine lag-probability map for a representative zebra finch recording. The distribution is broad rather than sharply peaked, consistent with sparse or ambiguous shared acoustic events. The learned branch nevertheless produces a smooth delay trajectory from the lag-domain representation. Peak-aware pooling summarizes both the location and dispersion of the lag responses for the temporal model, rather than selecting a single peak at each frame.

The best submitted system ranked first in the official BioDCASE

2026 Task 1 evaluation, with an average MAE of 309.68 ms across the two official test subsets. The validation results suggest that explicit lag-domain matching is particularly beneficial for coarse alignment, while the official test result confirms the overall effectiveness of the submitted hybrid system.

Table 2: Query-time ablation study of the neural delay estimator. MAE is reported in ms and MSE in s^2 . Average denotes the unweighted mean across the aru and zebra finch validation subsets. w/o indicates removal of the specified component, corr-only denotes a correlation-only cost volume.

Method	aru		zebra finch		Average	
	MAE ↓	MSE ↓	MAE ↓	MSE ↓	MAE ↓	MSE ↓
Full CV-Peak-BLSTM	118.4	0.020	422.9	0.262	270.6	0.141
w/o explicit cost volume	106.4	0.015	525.4	0.461	315.9	0.238
corr-only cost volume	105.6	0.015	518.4	0.424	312.0	0.220
w/o peak-aware pooling	106.1	0.015	497.5	0.403	301.8	0.209
coarse-only pooling	106.4	0.016	492.1	0.380	299.3	0.198
w/o BLSTM	106.0	0.016	480.5	0.382	293.2	0.199

Table 2 reports query-time ablations for the standalone neural estimator. Each model is trained once on the combined training set and evaluated separately on the aru and zebra finch validation subsets; the average is their unweighted mean. The full CV-Peak-BLSTM model achieves the lowest zebra finch and average MAE and MSE. Removing the explicit cost volume causes the largest degradation on zebra finch, increasing MAE from 422.9 ms to 525.4 ms. The correlation-only variant is also worse than the full model, supporting the complementary role of the negative absolute-difference term. Removing peak-aware pooling, using coarse-only pooling, or removing the BLSTM also increases both zebra finch metrics. The w/o BLSTM variant retains temporal convolutional context in the encoder and cost-volume refinement layers; its degradation is consistent with the benefit of longer-range recurrent modeling.

On aru, the neural ablations are close, with MAE values between 105.6 ms and 118.4 ms, and several variants slightly outperform the full neural branch. Thus, the benefit of the complete architecture is concentrated on the zebra finch coarse-alignment setting, while high-confidence aru recordings can be handled more precisely by GCC-PHAT.

The cost volume retains structured candidate-delay evidence, peak-aware pooling converts the lag responses into compact features and reliability descriptors, and the BLSTM integrates these features over time. Their combination is particularly beneficial when frame-level shared acoustic evidence is sparse or ambiguous.

6. CONCLUSION

We presented a peak-aware cost-volume model for multichannel bioacoustic alignment. The model adapts cost-volume matching to time-varying delay estimation and combines peak-aware pooling with BLSTM-based temporal modeling. Relative to the official neural baseline, the standalone model substantially reduces validation error in the zebra finch coarse-alignment setting. On aru, the standalone neural model does not improve over no synchronization; the fine-alignment gains of the hybrid system come from its GCC-PHAT branch. The final confidence-routed system ranked first in the official evaluation by routing high-confidence recordings to GCC-PHAT and lower-confidence recordings to the neural estimator. Evaluation is limited to BioDCASE 2026 Task 1. Future work will examine compact recording setups with smaller lags, cross-domain generalization, and adaptive routing.

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